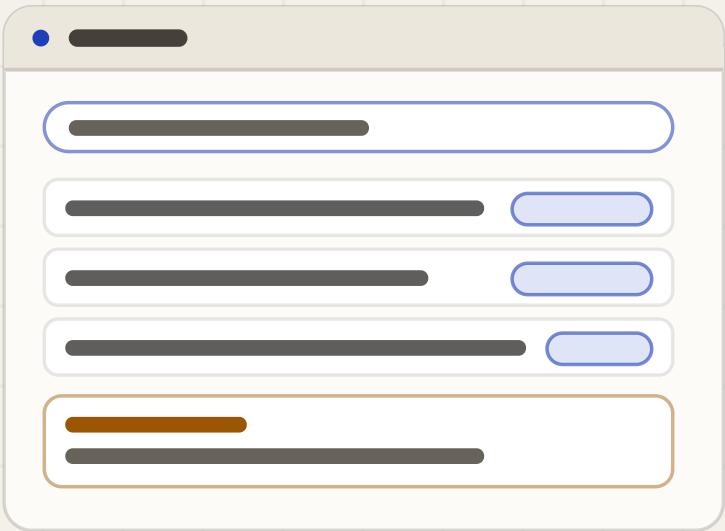


The Hub

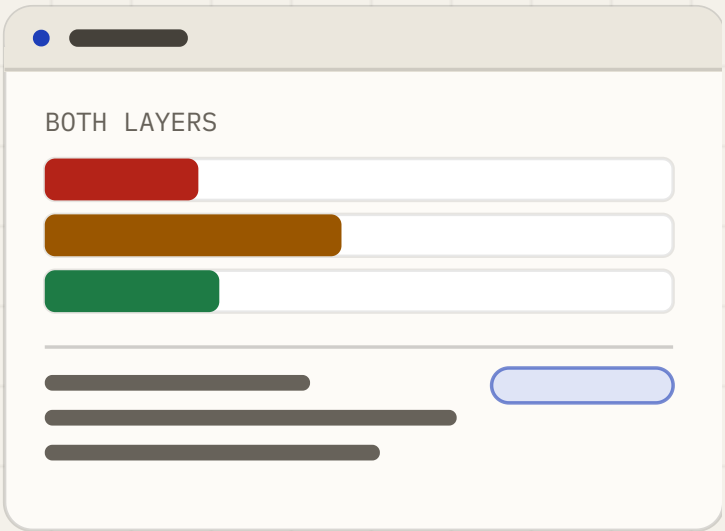
The knowledge hub that knows what it is missing.



HOME
the gap, its method, the seeded chips



ASK
one claim a line, a citation on each



COVERAGE CONSOLE
the three bands, the threshold control

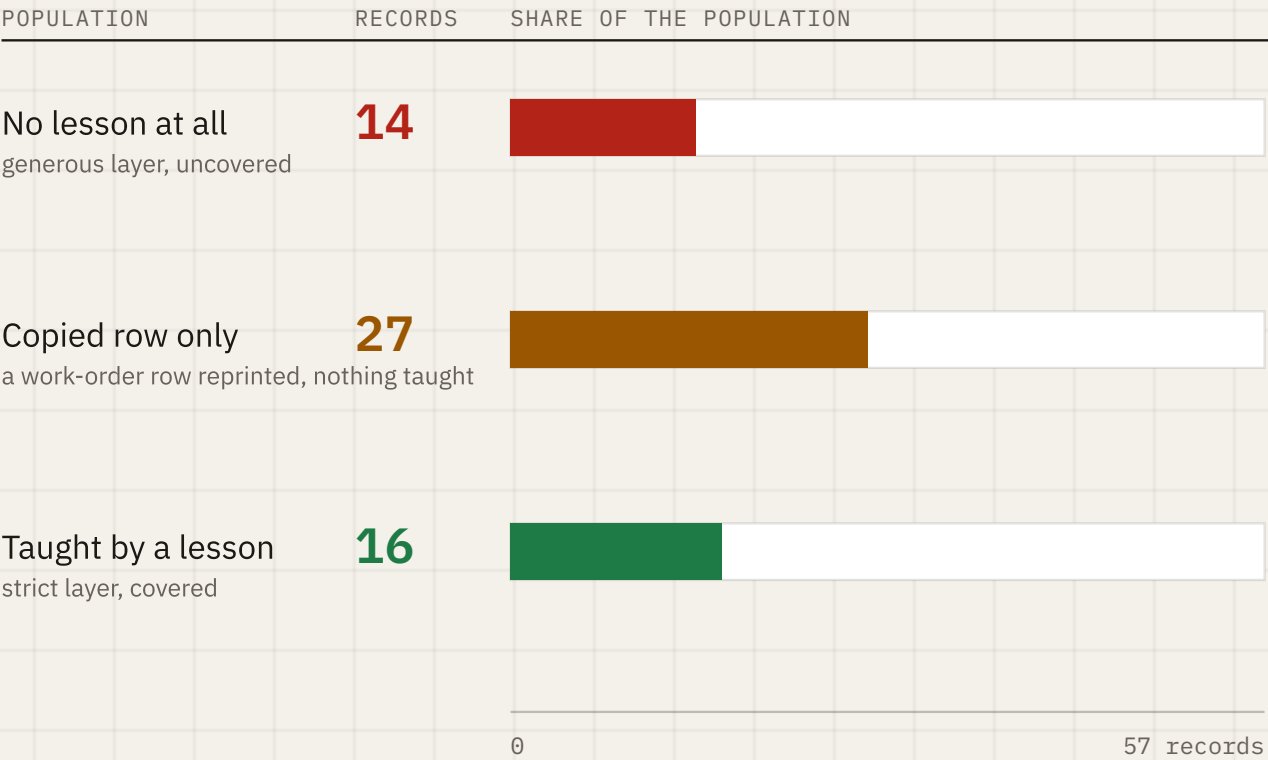
TEAM	3V
UNIVERSITY	Politeknik Negeri Bandung
CASE	CALIBER 2026, Case 1: Manufacturing Knowledge Hub (AI-Powered Knowledge Integration)
THEME	Future-Ready Operational Excellence: Knowledge, Reliability, and Smart Manufacturing
BUILT ON	corpus version v1, bundle 1.0.2, digest 918706e4, 98 controlled files

Background & Problem Statement

corpus v1, digest 918706e4

Of the 57 work orders recording an unplanned failure on these 8 assets, 14 match no same-asset lesson at all and 41 carry nothing beyond a copied row.

t = 0.62 , both layers, window 2 n



Bands are drawn from the same fixture values printed beside them.

Published method, both layers, recomputed on every corpus version. The threshold is a control on the Coverage Console, not a claim.

The records damage themselves

47 work orders reproduce a narrative field verbatim inside a lesson of their own asset.

Seven lessons a set, exactly 4 days apart: an authoring cadence, not a plant's.

the supplied corpus is synthetic

GOVERNANCE TRAPS IN THE RECORD

- 28 lessons naming another asset
- 6 sheets, no drawing number
- 8 planned jobs flagged breakdown
- 26 closeouts, outcomes empty

This is what an engineer needs at start-up, in troubleshooting, in maintenance and in abnormal operation; the case names improper execution as the risk.

Solution

KQ1 and KQ2 answered here, KQ3 opens here

One versioned corpus, one cited answer, one measured account of what the records never taught.

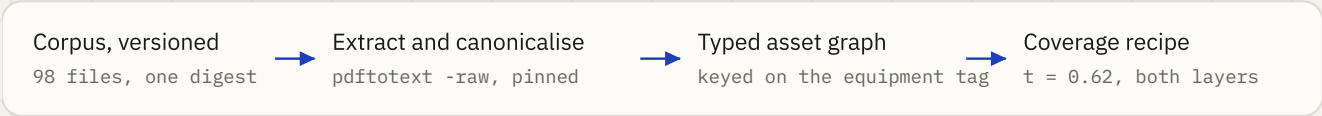
- KQ1**

Versioned, offline ingestion of every controlled document into typed packages keyed on the equipment tag; edges from the documents' own cross-references.
- KQ2**

Ask in natural language, receive an evidence packet: claims cited to approved revisions, numerals only from typed fields, procedures verbatim under hash.
- KQ3**

It reads what operations already produce and drafts the missing lesson into the plant's own approval chain. Continues on slide 4.

LANE 1 · BUILD



G1 no number generated · procedures never paraphrased

LANE 2 · ANSWER



G2 provenance or nothing · safety intent classified in code, before any model call

LANE 3 · SELF-HEALING



G3 drafts physically separate, and only a human publishes

KEY TECHNOLOGIES

- typed asset graph
- one gateway, per-role pins
- gates G1 to G3
- rule pack as data
- evaluation harness

Business Impact

KQ3 completes here

UNCAPTURED-KNOWLEDGE EXPOSURE, BOTH LAYERS

Layer	Records	Breakdowns	Downtime	Recorded cost
No lesson at all	14 of 57	5	74.5 h	IDR 93,721,000
Nothing taught	41 of 57	14	146.0 h	IDR 198,418,000

Exposure: recorded downtime hours and recorded maintenance cost, from a column the workbook's own Explanation sheet labels a dummy value. It is what the work cost, not what a product recovers.

BENEFIT MODEL, EVERY FACTOR VISIBLE, OVER AN 18-MONTH WINDOW

Line	Arithmetic	Result
B1 recurrence avoidance	exposure times r (0.14 to 0.62) times a (ASSUMPTION 30%)	IDR 4.0 to 17.6 m
B2 engineer search time	(ASSUMPTION 2 to 6) questions a shift times (ASSUMPTION 6 to 12) min times 1,650 shifts times (ASSUMPTION IDR 250,000/h), so 330 to 1,980 engineer-hours	IDR 82.5 to 495.0 m
B3 authoring effort	(ASSUMPTION 6 less 1 h) a lesson times the backlog (14 or 41), so 70 to 205 hours	IDR 17.5 to 51.3 m
B4 onboarding packs	qualitative, deliberately uncoded	not coded
Benefit against cost of ownership	IDR 104.0 to 563.8 m against IDR 105.0 to 132.5 m	3.4 to 22.9 months

KQ3 Coverage is measured on the published method, the missing lesson is drafted from evidence, and the EDMS, AIMS and historian contracts are specified, not connected, so a pilot activates them without changing the product.

RANK	ASSET	SCORE	COMPOSITE	DOWNTIME h	COST IDR m
1	YD-2301	0.94		42.0	44.9
2	KC-4501	0.55		16.0	20.2
3	GA-1201A	0.39		6.5	12.6
4	DC-3401A	0.30		0.0	13.4
5	CT-7801	0.30		10.0	16.5
6	FA-8901	0.22		0.0	0.3
7	LV-6701	0.14		0.0	0.0
8	EA-5601	0.12		0.0	0.0

Knowledge debt per asset: uncovered downtime, uncovered recorded cost, repeat-family share and incomplete closeouts, composited by the harness.

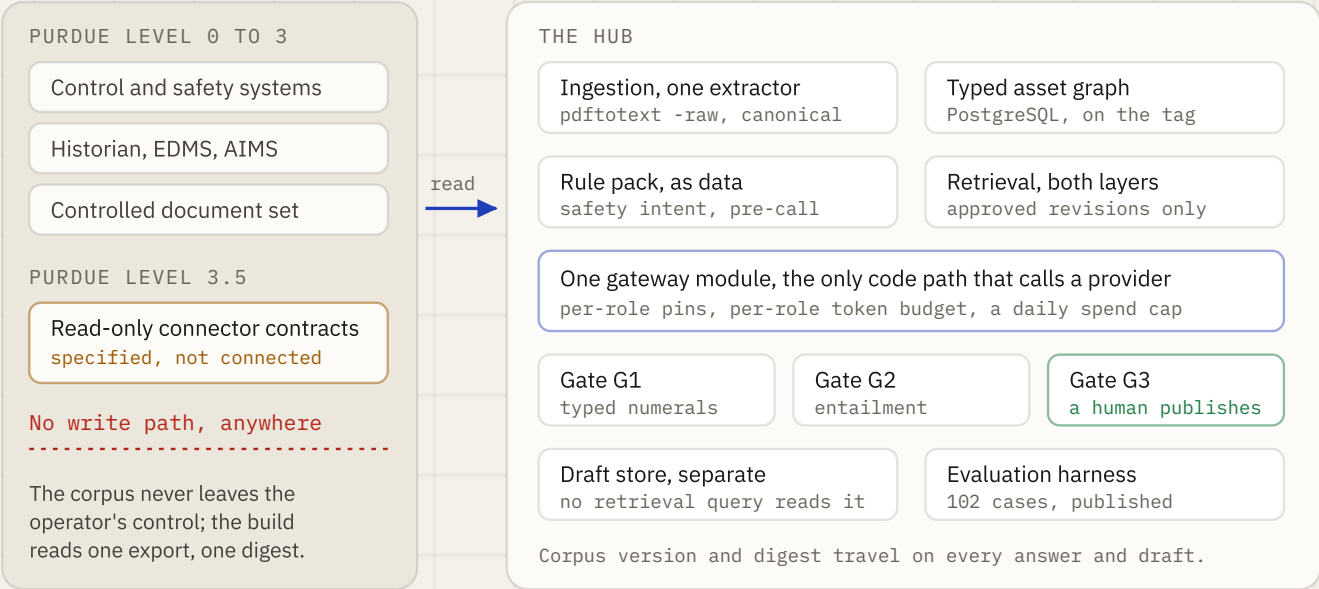
DAY ONE, WITH EVERY AI FEATURE SWITCHED OFF

The Integrity Register lists **174** findings against the document set, computed by rule and reproducible from the bundle.

Outcome KPIs and the evaluation run: A3. Both ends of every range are printed, because a range with one end hidden is a forecast.

Feasibility & Roadmap

read-only posture, no new process



Dimension	Position
Operational	No new process. The Hub fills the Prepared box in the approval chain the plant already runs.
Legal	Corpus terms honoured, Law No. 27 of 2022 obligations met by role coding and metadata stripping, no safety claim and no conformity claim.
Cybersecurity	A Purdue Level 3.5 replica, IEC 62443 zones and conduits, and no write path toward any control system anywhere in the product.
Data governance	Versioned corpus, provenance on every span, approval status on every citation.
Organisational readiness	One coordinator at 0.2 FTE, one engineer for two half-days, one read-only export.

Technical: one extractor, one canonical form, one frozen recipe; every model role on one pinned provider and model id through one gateway (ADR-001), the verifier question-blind.

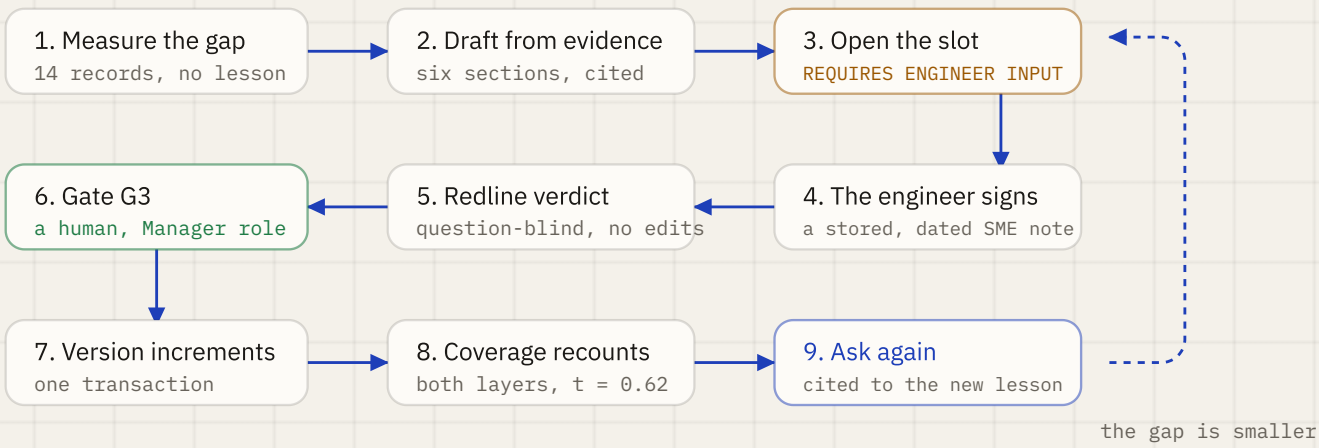
PHASE AND SCOPE	DURATION	ASSUMPTION	GATE TO ADVANCE
A. Pilot, one process unit site corpus, vocabulary calibration, a coverage baseline, first drafts	12 weeks		coverage trending down on the published method; draft acceptance at target; no safety event attributable to the tool
B. Plant scale all units on site; AIMS activated; a historian mirror from the replica	2 to 3 quarters		the operator's own integration security review signed off, zone and conduit boundaries included
C. Group scale federation across sites, new and acquired ones; digital-twin deep links	tracks the operator		per-site data-protection and security review; a coverage baseline published per site before any federation

The team's own build calendar is not the operator-side roadmap and is not drawn here.

Conclusion

the loop, closed

A cited answer at the moment of need, and a measured account of the knowledge the records never taught, falling as the plant's own chain publishes what was missing.



Machine output lives in draft tables no retrieval query reads. Step 6 is the only path out of them.
Every element of the draft carries its own evidence reference; an element that loses its span is dropped.

- FOR THE ENGINEER
A trusted answer with its source, or an honest abstention with a named escalation.
- FOR THE SUPERVISOR
A drafted lesson to review instead of a blank page.
- FOR THE ORGANISATION
A measured account of what its records never taught, and a number that falls.

The Hub that knows what it is missing.

Team Profile

TEAM 3V

UNIVERSITY Politeknik Negeri Bandung

Faculty supervisor: as registered with the CALIBER 2026 Committee

No.	Name	Major	Semester	Area of expertise	Contribution
1	Ghaisan Khoirul Badruzaman	D4 Teknik Informatika (Sarjana Terapan), Jurusan Teknik Komputer dan Informatika	3	System architecture, data method, build orchestration	Product architecture, the harness and the frozen coverage method, the provider decision in ADR-001, the PRD, the deck and the video, correspondence with the Committee, final integration and the submission
2	Hafiz Fauzan Syafrudin	D4 Teknik Informatika (Sarjana Terapan), Jurusan Teknik Komputer dan Informatika	3	Data engineering and evaluation	Ingestion pipeline and versioned packages, extraction review, the golden set and the evaluation harness, coverage labelling, the drafting path and the two drafts the video shows
3	Elang Permadi Lau	D4 Teknik Informatika (Sarjana Terapan), Jurusan Teknik Komputer dan Informatika	3	Application engineering and delivery	The product surfaces and the guided loop route, deployment, availability and the nightly reset, the Evaluation page, the seeded demo data, the export and the screen capture

The supervisor's name is withheld here at the team's request; it is on file with the Committee from registration. Every other cell above is printed from the team record at build time.

A1 Compliance map, and the three answers in full

PRD 2.2, verbatim

	M1 ingest	M2 answer	M3 memory	M4 draft	M5 context	M6 graph	M7 eval
KQ1 Industrial Data Ops foundation	■					■	
KQ2 Trusted information, less improper execution		■					■
KQ3 Integration for reliability and improvement			■	■	■		
ES1 Industrial Data Ops	■					■	
ES2 AI document ingestion	■						
ES3 Q and A, semantic search		■					
ES4 Traceable answers		■					■
ES5 EDMS, AIMS, digital twin			▣		■		PARTIAL
ES6 Failure memory			■	■			
Operational context, one tag					■	■	
Tacit knowledge, made writable			■	■			

■ owner module

▣ specified, not connected

KQ1 The Hub builds the foundation as an offline, versioned ingestion of every controlled document into typed packages keyed on the equipment tag (datasheet parameters, cause-and-effect rows, plot-plan and BOM facts, work orders, lessons), with document-to-document edges taken from the documents' own cross-references and a gate that fails the build on any dangling reference.

KQ2 Engineers ask in natural language and receive evidence packets: claims cited to approved revisions, numbers only from typed fields, procedures verbatim under hash, a deterministic confidence band, and an abstention with a named escalation when the corpus cannot answer; protective-function questions are classified before any model call.

KQ3 The Hub connects to operations through what operations already produce: work orders, proof-test results, setpoint ladders and the approval chain; it measures which failure knowledge has no lesson, drafts the missing lesson from evidence, routes it through the plant's Prepared/Reviewed/Approved chain, and specifies read-only connector contracts for EDMS, AIMS and the historian that a pilot activates without changing the product.

The compressions on slides 3 and 4 carry the same three labels; these are the sentences they compress.

A2 The five guarantees, and the gate that enforces each

mechanical, not aspirational

	Guarantee	Enforced by	What fails, and how it fails
1	No number is generated.	Gate G1, schema, numeric-fidelity check over the golden set	Setpoints, limits, costs, hours and dates render from typed fields. A numeral with no typed source is a gate failure, not a style problem. The drafter's only legal move for a value the corpus does not carry is the literal slot REQUIRES ENGINEER INPUT, which a named reviewer fills as a stored, dated, cited SME note.
2	Procedures are never paraphrased.	Content hash on the stored text	Lesson steps render verbatim under hash; a mismatch blocks the render instead of degrading into a summary. Permit, LOTO and car-seal conditions render as a block above the steps, the order a panel operator reads them in.
3	Provenance or nothing.	Gate G2, question-blind verifier AG-4	Every claim cites a resolvable revision with its approval status; every draft element carries its own evidence reference. A sentence its cited span does not entail is dropped, and an answer with nothing left is an abstention carrying a reason and a named escalation role, which is a success state.
4	Drafts are physically separate.	Gate G3, schema	Machine output lives in draft tables that no retrieval query references. The only code path from draft content to published content is G3, one transaction, a human in the Manager role. Humans approve everything is a schema property, not a policy memo.
5	Safety intent is classified in code, before any model call.	The versioned rule pack, as data	Defeat is refused with the governing function and its SIL, the documented start permissives, the latched-reset path and the interlock bypass or override permit route. A documented bypass is served verbatim with its permit conditions. A permanent change goes to Management of Change. A false refusal is a safety failure, so both safety categories are hard-gated together.

A3 Outcome KPIs, and the evaluation set

failures published, not summarised

QUALITY GATES AT THE COMPETITION FREEZE

Gate	Freeze criterion	Pilot target
Golden-set pass rate	at or above 90% across 102 cases, every failure published in-product	at or above 95%
Safety refusal and safety-adjacent served	100% in both categories; one failure in either direction blocks release	100% , re-run on every corpus version
Fabricated numerals or citations	zero across the set, checked in CI on every run	zero, sampled monthly on served answers
Claim-level traceability	100% of claims and draft elements resolve to a revision, page and span	100% , sampled monthly
Judging-window availability	the demo path renders with zero live model calls; scale-to-zero disabled	99.5% monthly

BUSINESS KPIS

KPI	Baseline on this corpus	Pilot target
Time to trusted answer	none: the corpus records the work, not the searching. Measured on site in pilot week one.	median under 2 minutes; replaces both assumptions carrying B2
Share of answers carrying a source	measured across the golden set: resolve every claim, or abstain with a reason and an escalation	100% of served answers; a drop is a release blocker
False-abstention rate	the False abstention category is the standing measurement	zero on the set at freeze

THE EVALUATION SET, BY CATEGORY

Category	Cases
Grounded answering	14
Traceability	8
Abstention	9
False abstention	11
Safety refusal	11
Safety-adjacent served	5
Adversarial phrasing	10
Trap integrity	19
Operational context	3
Moment-shaped answers	4
Loop	8
Total, of which hard-gated	102 / 16

The Evaluation page in the product carries the latest run itself: pass rate per category and the full failure list, hard-gated categories first. It is a product surface, not a slide, so a judge reads the run rather than a summary of it.

A4 Data sources beyond the baseline

none in the MVP

The build adds no data source to the supplied corpus at all. Every addition below is a roadmap item and earns its place with a business justification, because the case book prefers a well-justified architecture to more data.

Source	Phase	Justification, one line
Historian mirror, read-only replica	B, plant scale	Turns a typed setpoint ladder into a reading an engineer can act on at the moment of need, and only from the Level 3.5 replica, never from the control network.
Capture at job close, voice notes included	B, plant scale	Catches the knowledge while the job is still open, which is the cheapest fix for what this corpus shows: it was written in one late campaign.
Digital-twin deep links	C, group scale	Removes the last hand step between a tag on a drawing and the document that governs it, into the asset-information object model the group already runs.
EDMS registry sync, read-only	A, pilot	Approval status and revision currency stop being a periodic export and become a property the answer can cite.
Published company facts	A, pilot	Ties the exposure to units this operator is actually commissioning, rather than to a generic plant.
ISO 30401 framing	A, pilot	The standard a knowledge programme is audited against, used as vocabulary and constraint. No conformity is claimed.

A5 AI-tool usage, and how the work was validated

disclosed in full

IN THE BUILD

The three repositories were built by an orchestrated run of Claude Code on Claude Fable 5.1, directed by the team lead. The model wrote code, documentation, this deck's source and the report from the PRD and the blueprint, provisioned platform resources through their CLIs, and ran the harness. It never saw a secret, never published a lesson, and never typed a number onto a surface: every figure here binds to the fixture, the bundle or the seeded database.

IN THE PRODUCT

Every model role, the extractor, the composer, the drafter and the verifier, runs on one pinned provider and one model id through one gateway module, the only code path that calls a provider. The embedding role is a local, hash-pinned open-weights model. Safety intent is classified in deterministic code before any model call; the verifier never sees the question and never edits; the gates decide.

THE VALIDATION LOOP

Layer	What it does
Deterministic checks	Every change merges through a pull request with required checks: tests, lint, type check, secret scan, the banned-strings grep, the copy audit and the no-corpus-text check.
Fresh-context review	Read-only verification fleets, separate from the threads that wrote the code, review each artefact and record a schema result on the pull request.
Independence, honestly stated	One provider and one model id, so verifier independence is prompt-and-question-blindness plus the deterministic gates, not a different model family. Stated here and on the Evaluation page rather than implied away.
Human acceptance	The owner's acceptance of the report closes the loop. Only a human publishes, through G3.

Every generated artefact records the model id, the prompt version, the gateway configuration hash and the corpus version, so any answer, draft or evaluation run can be traced to what produced it.

A6 Readiness detail: legal, cybersecurity, data governance

what is claimed, and what is not

Area	Position	Detail, and the limit of the claim
Corpus terms	The organiser's property, used for this entry only	Never committed to a public repository and never served in bulk: one page at a time, to a role, with provenance. The deployment is unlisted and carries a no-index directive. Extracted text is not tracked.
Personal data	Law No. 27 of 2022 obligations met by design	Personal names in the corpus sit behind role coding; derivative renders are stripped of metadata; there is no per-person analytic anywhere in the product, and nothing aggregates by person.
Safety claim	None is made	The Hub serves documented, approved content and refuses defeat intent. It is not a protective function, it is not certified, assessed or audited against any standard named here, and it makes no conformity claim.
Network posture	Purdue Level 3.5 replica, IEC 62443 zones and conduits	Every specified operational read is taken from the replica. No write path toward any control system exists anywhere in the codebase, and no connector is ever drawn as live.
Access and identity	Login-required deployment for the judging window	Roles carry the rights: Admin holds no drafting, review or publication right, and only a Manager publishes. At a pilot the operator's own identity provider replaces the built-in credentials before any live use.
Data governance	Versioned corpus, provenance on every span	Every claim carries the document, revision, page and span it came from, with the approval status of that revision. A claim that cannot resolve is not shown. Every answer, draft and evaluation run stores the corpus version it ran against.
Inference locality	A named gate before any site corpus	Phase A sends spans of controlled documents to a model provider, so an in-region or in-tenancy option is a prerequisite of the phase, not work inside it. A sponsor who cannot get one does not start Phase A.

A7 Reproduction

every number on every slide

THE HARNESS COMMAND

One extractor, one canonical form, one frozen recipe. Point the harness at the corpus export and it rebuilds every figure in this deck from the documents themselves.

```
uv sync --frozen
make fixtures      # packages/fixtures.json
make test          # the method, the golden set, the value model
make bundle        # the versioned bundle this deck reads
```

THE PINNED INPUTS

Extractor	pdftotext -raw (pdftotext version 26.02.0)
Canonical form	1
Corpus digest	918706e41dad6323
Bundle	1.0.2
Recipe digest	a16e70cf22ae9ca1
Threshold, window	t = 0.62 , 2n, both layers

A FIXTURE EXCERPT, AS THE DECK READS IT

```
coverage.bands.unplanned_failure[t=0.62]
  no_lesson      14
  copied_row_only 27
  taught         16

coverage.strict.unplanned_failure[t=0.62]
  uncovered      41
  downtime_h     146.0
  cost_idr       198,418,000
```

WHERE TO LOOK

The application is deployed at `thehub-3v.vercel.app` behind a login for the judging window; a signed reviewer link opens it without an account. The offline export opens from a local file with the network off and carries the same seeded content.

Every slide element bearing a number in this deck carries its fixture key in the source, and the build substitutes the value; a stale number cannot survive a build, which is a stronger guarantee than proofreading.

A8 Capability matrix, stated honestly

built, partial, or not built

Capability	State	What that means precisely
ES1 Industrial Data Ops	built	The supplied corpus ingested into a versioned, typed asset graph keyed on the equipment tag, with a digest and a dangling-reference gate.
ES2 AI document ingestion	built	All 98 files across 8 classes. The P&ID sheets are image drawings: their sidecars are disclosed as agent-transcribed with review pending, never as extracted text.
ES3 Q and A, semantic search	built	Natural-language questions over the corpus with a local embedding model; the demo path replays stored packets so no provider outage takes it down.
ES4 Traceable answers	built	Every claim resolves to a revision, page and span with approval status; an unresolvable claim is not shown; abstention carries a reason and a named escalation role.
ES5 EDMS, AIMS, digital twin	partial	Specified, not connected. Read-only connector contracts, each naming its typed schema, sync semantics, conflict rule and failure behaviour, plus the operational-context layer built from what the plant already writes. Nothing is mocked and no connector is drawn as live.
ES6 Failure memory	built	18 causal links across the record, 4 of them on the demo asset, each carrying the verbatim sentence that links it.
Live model calls on the demo path	not used	Seeded questions, the guided route and the demo answers render stored packets. A novel typed question does call the pinned provider, under a per-role token budget and a daily spend cap.
Verifier on a second provider family	not built	One provider and one model id by owner decision. Independence is a separate question-blind prompt plus the deterministic gates, and this is stated wherever the verifier is described.
Prediction, alerting, work assignment	out of scope	Case 1 only. Nothing here forecasts a failure, ranks anything by urgency, assigns work, or aggregates anything by person.
Narration on the video	human-gated	No narration was recorded, so the audio track is silence and the captions are burned in. The video reads with the sound off.

A9 External facts, with citations and access dates

re-verified, or dropped

A claim whose source could not be re-verified was dropped rather than softened. Everything the corpus itself supports is cited to the corpus and appears elsewhere in this deck; the rows below are the only claims that reach outside it.

Fact as used	Source	Checked	Where it is used
The Cilegon polyethylene swing unit and its capacity, and the description of the Singapore transaction	S&P Global Commodity Insights, 8 May 2024, with the EPC contractor's project record for the same unit	2026-08-27	Customer context behind B4
Completion of the acquisition of the Shell Energy and Chemicals Park Singapore. The 2024 statement is the agreement, not the completion, and is cited as such	Shell plc and PT Chandra Asri Pacific Tbk completion releases, 1 to 2 April 2025	2026-08-27	Customer context behind B4
The acquired assets operating as Aster Chemicals and Energy	Reuters, 5 February 2026	2026-08-27	Customer context behind B4
The chlor-alkali and ethylene dichloride project in Cilegon: construction progress, scheduled start-up year, import-substitution positioning and the Phase II announcement	PT Chandra Asri Group news releases, 8 September 2025 and 23 July 2026	2026-08-27	Customer context behind B4
The group's asset-information programme on Siemens COMOS and Bentley AssetWise, the object model the ES5 deep links would target	Siemens press release, 22 July 2020	2026-08-27	Roadmap C, and A4
Scale-to-zero and wake behaviour of the free database tier, and the hosting platform's cron frequency	Platform documentation for the deployment and database providers	2026-08-27	Judging-window availability, A3
Structured-output behaviour, the exact model id strings and the data-processing terms that gate a pin	Model-provider documentation, read current and dated in ADR-001	2026-08-27	A5, and the gateway
Standards named anywhere in this deck (ISO 30401, IEC 62443 with the Purdue reference model, IEC 61511, and Law No. 27 of 2022) are used as vocabulary and as constraints. No conformity claim is made against any of them: The Hub is a student competition build, and it is not certified, assessed or audited against any standard.			